

Global Industrials

Datacenter 800V DC – Differentiated take on SemiAnalysis industry primer

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SemiAnalysis published a detailed technology deep-dive on the transition to 800V DC on 26 May 2026, with a free version available on its website. Given the firm's reputation and the depth of the analysis, investors may view the report as an important reference document. While we agree with many of the report's conclusions, we also highlight a number of differentiated perspectives and set out a broader framework for assessing the implications for datacenter electrical equipment suppliers, including Schneider Electric, Vertiv, Eaton, ABB, Legrand and Siemens. In particular, we seek to map the potential impact of the technology transition to each vendor's specific areas of exposure, although this analysis is inevitably constrained by the limited level of disclosure.

In summary, the majority of datacenter revenues for most electrical equipment vendors are likely to go through a technology transition over the next five-plus years, creating both uncertainty and new opportunities. This is not only a "LeGrand and rack PDU" issue, which has become a common early mis-characterization. It is a broader transition affecting the sector's low-voltage electrical, LV transformers, power electronics, UPS, distribution and cooling profit pools. We believe that there is significant nuance in terms of how this may affect different portfolios, both positively and negatively. Simply thinking in terms of binary outcomes is much too simplistic.

General conclusions on 800V DC:

- **Higher-voltage DC is becoming a physics-led requirement, not a design choice:** The move towards a higher-voltage direct current architecture for AI datacenter power supply over the next five years is increasingly being driven by physics, given the power densities expected in next-generation compute. 800V DC – a higher-voltage "low-voltage" architecture rather than a high-voltage system – appears set to become the preferred architecture, although even higher voltages may be required over the longer term.
- **The objective is to power more compute within a denser layout with lower infrastructure cost:** The ultimate goal of these new architectures is to enable more powerful compute capacity in a dense setting to be powered with lower capex and opex in the supporting electrical infrastructure. This is generally negative for electrical supplier growth relative to the underlying growth in gigawatts of compute demand, as each MW of compute may require less traditional electrical equipment than would have been the case with today's architecture.
- **The market shifts from grey-space AC equipment towards white-space power electronics:** Under an 800V DC architecture, the electrical vendor opportunity at the medium-voltage level remains largely intact. However, the grey-space low-voltage market transitions from AC to DC, reducing the role of traditional LV UPS, switchgear, transformers and PDUs to auxiliary facility-level power loads and cooling. At the same time, the share of spending captured by white-space power electronics increases.
- **The total opportunity per MW changes only modestly, but the mix changes significantly:** According to the SemiAnalysis framework, the addressable market for electrical suppliers falls from c.\$3.35m per MW of compute in the baseline scenario, excluding generators, to c.\$3.05m per MW in Phase 4. This reduction is modest relative to the expected underlying growth in compute demand. However, the composition of spend changes dramatically: c.\$1.25m of the Phase 4 opportunity is represented by the emerging solid-state transformer market, a new technology category without established market leaders, where start-ups and transformer manufacturers such as GE Vernova and Siemens Energy may compete with traditional DC electrical equipment vendors. The addressable market in Phase 1 expands, with only rack PDUs impacted.

Oxcap's points of differentiation versus the SemiAnalysis framework:

- We expect the transition to be more fluid than the clearly defined four-phase framework outlined by SemiAnalysis, which is heavily anchored around the NVIDIA roadmap. ASIC-based chip designs, as well as architectures from AMD and Chinese vendors, may follow different paths. Datacenter power architectures are already heterogeneous across hyperscalers and other operators, and we expect this to remain the case.
- While it is clear that the traditional AC low-voltage opportunity is likely to shrink materially, DC architectures will also require control, protection and monitoring. This could create additional opportunities that may not be fully captured in the SemiAnalysis framework. Again, we believe that AC and DC outcomes are not binary.
- Technology position is only one factor in vendor selection. The ability to manufacture at scale, maintain consistently high product quality and meet tight delivery schedules is critical. These capabilities, combined with close customer relationships, represent an important advantage for incumbent electrical suppliers, particularly relative to new entrants.
- Hyperscalers will also want to maintain a diversified supplier base and are unlikely to push a major technology transition before a sufficiently broad set of qualified vendors are ready. The portfolios of electrical suppliers are not static, and we expect further R&D investments, M&A, partnerships and technology licensing as vendors position themselves for the evolving roadmap. We therefore disagree with the idea that there are already clearly defined winners and losers, particularly given that the technology roadmap beyond Phase I is not yet set in stone.
- We address some of SemiAnalysis' stock conclusions in more detail below. In simple terms, the report characterizes Delta Electronics and Vertiv as relative winners; ABB, Eaton and Schneider as somewhere in between; and Legrand as the least well-positioned (with Siemens not discussed at all). Although we understand these directional views, this does seem like an over-simplification. The analysis focuses mainly on near-term effects (eg. rack PDU, low-voltage UPS) but does not capture potential growth in other categories. Nor does it really address different companies' ability to pivot their businesses. While the share of revenues affected by the design transition is highest at Legrand, the exposure is also material for ABB, Siemens, Schneider Electric and Eaton, and to a lesser degree Vertiv. Although cooling is not the focus of the SemiAnalysis study, that market is also undergoing significant change, particularly with the move towards direct-to-chip liquid cooling with current designs likely to evolve again.
- From a bigger picture perspective, the shift of spending from traditional AC electro-mechanical content to DC and power electronics represents a challenge for our group of suppliers which built very successful and fast growing datacenter supply businesses on top of their traditional electrical offerings in relatively stable and consolidated markets. The transition will take time and incumbents have every right to win if they execute well. There is however a displacement risk that should not be underestimated.

Technology transitions always represent a risk, and could coincide with a growth normalisation

- **Pricing conditions may change:** To date, the AI datacenter power opportunity has largely accrued to established electrical suppliers, often through their existing product portfolios in relatively consolidated markets. Strong datacenter demand has tightened capacity and supported pricing power, benefiting their broader medium- and low-voltage power businesses and contributing to material margin expansion. As datacenters transition towards DC architectures for the lower-voltage power system, pricing conditions in traditional AC markets could begin to change.
- **Shift from electromechanical to power electronics – margin questions:** The shift towards higher power-electronics content, and away from more traditional electro-mechanical equipment, may also be negative for margin mix. It could increase the share of outsourced supply and expose incumbents to greater competition, particularly from Asian vendors with strong positions in power electronics.
- **Growth effects not visible until 2028–29:** In the near term, Phase I of the transition appears broadly additive to the vendor opportunity, with the exception of the rack PDU market, only relevant to Legrand in terms of size of exposure, while underlying growth in compute demand is expected to remain strong. As a result, we would expect the technology-driven impact on demand and vendor growth to become more visible only from around 2028 or 2029.
- **Hyperscale capex – important variable:** However, if this technology transition coincides with a moderation in hyperscaler capex growth, the impact of shifting demand patterns could become much more material for individual vendor growth rates by then.

Vendor impact - our assessment

SemiAnalysis concludes that **Delta Electronics** is best positioned for the coming design changes. We agree with this assessment. Delta is a leader in power electronics, has strong DC capabilities and has less to lose from disruption to the traditional AC architecture.

SemiAnalysis sees the greatest risk at **Legrand**, reflecting the near-term negative impact from a move away from rack PDUs and a weaker position in emerging DC markets. Legrand disagrees with this conclusion and has referred investors to its 29 September 2026 CMD for more detail on how it intends to position for the transition, including new DC products. Investor concerns around the 800V DC transition have so far focused heavily on Legrand and its PDU business, meaning this risk may already be at least partly reflected in expectations.

SemiAnalysis also views **Vertiv** as well positioned, given its lower relative exposure to categories most negatively affected by the transition. This is supported by the large share of cooling in its business mix and the group's position in electronics-rich infrastructure.

ABB and **Siemens** have a relatively high share of datacenter exposure in medium-voltage products, which should not be materially affected by the transition, and a lower share of exposure to UPS, a category that could be phased out over time. In Siemens' case there is no UPS exposure. ABB also has a strong position in DC technologies, in our view, supported by its heritage in DC applications such as Marine and EV charging. However, we would also highlight the potentially underestimated exposure of both ABB and Siemens to general low-voltage electrical power solutions in datacenters, parts of which may either disappear or need to transition from AC to DC.

SemiAnalysis concludes that **Schneider Electric** and **Eaton** are less favourably positioned than Vertiv and Delta for the 800V DC transition. We believe this is driven more by exposure mix than by an assessment of technological capability. UPS / Secure Power Solutions are central to the datacenter offering of both Schneider and Eaton, accounting for around one-third of their datacenter sales, and could be at risk as architectures evolve. That said, both companies are likely to become key suppliers of the incremental DC architecture, including HVDC power racks, battery racks and related infrastructure. Eaton's recent acquisitions of Boyd, in cooling, and Resilient Power Systems, in solid-state transformers, also changed its exposure map and should be considered in any assessment of its future positioning.

NVIDIA's own 800V DC materials ([Link](#)) list ABB, Eaton, Schneider Electric, Siemens and Vertiv as electrical ecosystem partners, but do not appear to include Legrand in the same 800V DC partner roster. This likely reflects a difference in offering: Legrand provides a set of important components, including rack PDUs, busway, cabinets and connectivity, but is less clearly positioned as an end-to-end electrical systems supplier in the same way as some of its peers.

When assessing the risks to **Legrand**, it is also important to recognise that the rack PDU concern is already discussed among investors. Legrand has built a highly successful datacenter business over the past 8–10 years, supported by M&A, and we would expect the company to adapt its portfolio as the market evolves, including through further acquisitions. In addition, Legrand's product categories have not benefited from the same level of pricing and margin expansion seen in areas such as medium voltage, and likely UPS, where tight supply-demand conditions have lifted profitability for some peers well above historical levels. Those profit pools may be more exposed if market conditions normalise.

While the share of revenues affected by the design transition is highest at Legrand, the exposure is also material for ABB, Siemens, Schneider Electric and Eaton, and to a lesser degree Vertiv. Although cooling is not the focus of this report or of the SemiAnalysis study, that market is also undergoing significant change, particularly with the move towards direct-to-chip liquid cooling. This creates risks for Vertiv's traditional cooling offering, but also incremental opportunities in emerging cooling technologies for Vertiv, Legrand, Eaton and Schneider Electric. ABB and Siemens have limited, if any, exposure to cooling. The currently envisaged two-phase liquid cooling architecture and the use of CDUs could themselves be challenged over time by single-phase liquid cooling solutions that do not require a CDU or new cooling technologies (immersion cooling for example) while traditional AC-powered CDUs may be replaced by DC-powered CDUs.

SemiAnalysis Vendor assessment: SemiAnalysis concludes that the move to 800V DC shifts spending towards whitespace electronics and highlights **Delta Electronics** and **Lite-On** as the main beneficiaries. It also believes that these and other electronics vendors may compete for business with the traditional datacenter electrical equipment suppliers of the AC area. Among the traditional vendors it sees **Vertiv** as best positioned given its electronics capabilities and strong position in liquid cooling, resulting in an overall more attractive mix with lower % exposure to AC categories getting eliminated. It states that it sees **Schneider** as structurally behind Delta and Vertiv in the 800V DC race but it notes their strong position in MV which is not affected by the technology change. It notes that **Eaton** has a smaller exposure to white space areas it expects to grow faster. It notes Eaton's early move into solid state transformers. **ABB** is seen as having a narrower offering than Vertiv, Schneider and Eaton. **Siemens** is not discussed in detail, potentially a reflection of the group's narrower exposure to MV and LV electrical products but maybe underestimating the size of their datacenter business (similar to ABB and Legrand). **Legrand** is seen as most at risk given its high share of datacenter revenues exposed to AC low voltage power including busbars, busways and rack PDUs, which are already negatively impacted in Phase 1 of the transition. It concludes that Legrand is less well positioned to capture incremental opportunities in DC solutions such as the sidecar.

When analysing the datacenter technology outlook, Oxcap applies a multi-layered framework:

1. **Power Consumption vs. Unit Volume:** A key question is whether equipment demand is driven primarily by power consumed or by the number of physical units deployed. As rack power density increases, the number of racks – and therefore the number of rack-level components – grows more slowly than the power required to operate the system. This favours categories that scale with megawatts: power generation, medium-voltage equipment, secure power and cooling. This contrasts with categories that scale more closely with rack count: cabinets, cable management and certain types of PDUs.
2. **Equipment Lifespan:** An important differentiator, as it determines the balance between replacement capex and growth capex. Transmission equipment and large transformers can remain in service for 40 years or more; power generation equipment typically has a 20–30-year useful life; broader electrical equipment may last 10–15 years; while equipment closest to the server – including parts of the cooling and rack-level power chain – may be replaced or redesigned with each compute upgrade cycle, potentially every 4–6 years. As the current breakneck pace of AI capex eventually normalises, replacement-cycle exposure should become a more important driver of relative growth.
3. **Technology Transition:** Investors need to assess how architectural shifts, such as 800V DC, liquid cooling and higher rack densities, affect demand by product category and alter the competitive landscape within each category. Mature, oligopolistic markets often deliver the highest margins and most stable returns. By contrast, technology transitions can open the door to new entrants, shift value pools, and require higher upfront R&D, qualification and product ramp costs before margins normalise. The change from electromechanical content toward electronics and semiconductor-based technologies can have negative implications for existing vendors, also in terms of the profitability profile. Electronics technologies often experience faster price deflation, a higher share of outsourced components and a more fragmented competitive landscape, with Asian suppliers potentially playing a much larger role.

Our view on 800V DC transition and the SemiAnalysis framework

- **A "Fluid Transition":** The SemiAnalysis framework describes a four-phase transition towards a full DC, grid-to-chip power architecture by the end of the decade. In practice, we expect this transition to be more fluid, with different customers adopting different design solutions depending on workload, chip architecture, power density, facility design and supplier ecosystem. We also expect the traditional AC datacenter to remain an option long term for certain applications. There is a fair question whether standard inference will need large, superdense datacenters longer term, particularly given the advances in condensed models that will be good enough for many tasks and can be run locally, either on premise or in smaller traditional datacenter.
- **Multiple Design Approaches:** We would also note that, while the SemiAnalysis framework is anchored around NVIDIA's roadmap, ASIC-based architectures from Google and Amazon may follow somewhat different design paths. The same is likely true for datacenters built around AMD accelerators or Chinese chip architectures. Multiple design approaches for the electrical system will likely persist at the same time.
- **Higher Power Density = Higher-Voltage DC Architecture:** Given the pace of technological change, ongoing efficiency improvements and the still-evolving use cases for AI, it is difficult to predict with confidence what the AI infrastructure landscape will look like even 12 months from now, let alone by 2030. What does seem clear is that power density will continue to rise. This is likely to force a move towards higher-voltage DC power architectures, away from today's standard low-voltage AC approach, which is based on building and industrial electrical technologies that have been used for decades.
- **DC also needs control and protection:** DC is a complex and potentially dangerous technology. The implementation of a full DC technology in Phase 3 and 4 is likely more challenging than what is implied by just drawing a line in a design chart from the solid state transformer to the battery pack next to the rack, representing an 800V DC copper busway. The SemiAnalysis framework may underrepresent this opportunity. DC is inherently more challenging to manage than AC due to the lack of a natural current zero crossing. Fault currents rise extremely quickly and electronics on either end of the busway are very vulnerable, hence the protection required is likely higher than for AC systems, an incremental opportunity for our suppliers, also from a technology differentiation and price perspective. It will require DC / solid state circuit breakers at either end, protection relays, sensors, monitoring devices, arc flash detection and battery management systems. These are all incremental opportunities for today's low voltage electrical suppliers if they invest and make the technology transition.
- **Partnerships Matter:** A vendor's technology position and product offering are only part of the decision-making process. Hyperscalers also care about maintaining a competitive supplier landscape and are therefore keen to work with multiple vendors, while introducing design changes at a pace that avoids creating monopolistic or oligopolistic supplier structures. The large number of partnerships between electrical suppliers and NVIDIA is evidence of this co-design approach.
- **M&A Potential:** Hyperscalers also place significant weight on a supplier's ability to deliver large volumes on tight timelines, while maintaining consistently high product quality. A start-up may offer differentiated technology, but that does not necessarily mean an incumbent supplier will be displaced. Instead, we would expect existing suppliers to respond also through M&A, partnerships and technology licensing as DC power architectures evolve. For example, we would not be surprised to see additional acquisitions of, or partnerships with, emerging solid-state transformer suppliers.

Vendor exposure map

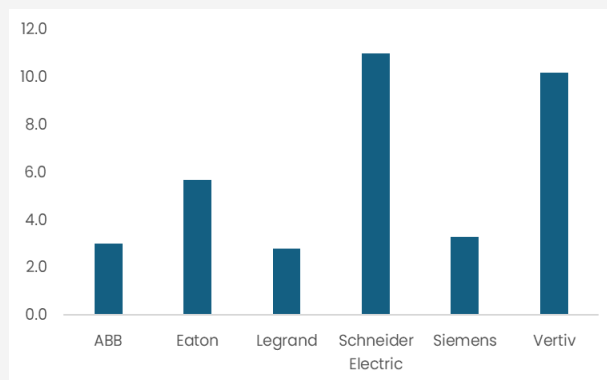
Company disclosure on their datacenter revenues and offering by type is limited. We encourage companies to step up disclosure, now that for many, datacenters are not just the fastest growing market, but also represent a sizeable share of revenues. As the technology transitions, and the break-neck speed of growth eventually starts to normalize, investors will look for more information. The combination of the usual quarterly volatility with investor uncertainty around the technology roadmap may well result in more volatile share prices. We have already seen this in the last two years, where companies are disproportionately punished, or rewarded, on data where there is too much guesswork and backing out. The situation could become far worse, as growth normalizes.

The chart below represents our best effort to split 2025 revenues of the vendors under coverage into standardised categories. Schneider, Legrand, Siemens and Vertiv provide some disclosure by offering but each use their own categories, often bundling groups together. Hence Figure 3 below likely has inconsistencies and mistakes and we are keen to get feedback.

We apply the following breakdown:

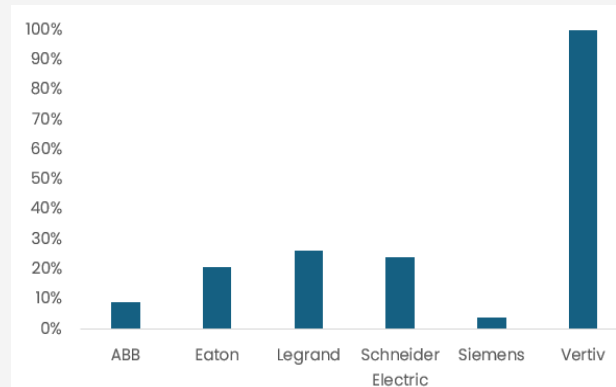
- **Medium Voltage:** Medium voltage switchgear and transformers (less relevant as mostly supplied by other companies including Hitachi, Siemens Energy, GE Vernova).
- **Low Voltage Electrical:** Low voltage switchgear, LV transformers, protection equipment, busways.
- **Uninterruptible Power Supply (UPS):** Medium and low voltage UPS systems including batteries.
- **Rack PDUs:** Power strip/distribution device taking power from the upstream electrical system and distributes it to the IT equipment in the rack.
- **Cooling:** Solutions including traditional in-row or in-rack cooling, chillers but also liquid cooling solutions including CDUs. Vertiv has a large traditional (and liquid) cooling franchise while for the rest of the coverage, it's targeted on specific applications.
- **DC Infrastructure:** Racks, cable management, installation products.
- **Building Control:** Building management systems, HVAC, fire and security.
- **Software & Services:** Maintenance, datacenter infrastructure management software.
- **Distributed IT:** Single phase UPS, secure power not related to datacenters.

Figure 1: Datacenter revenues 2025 in US\$ bn



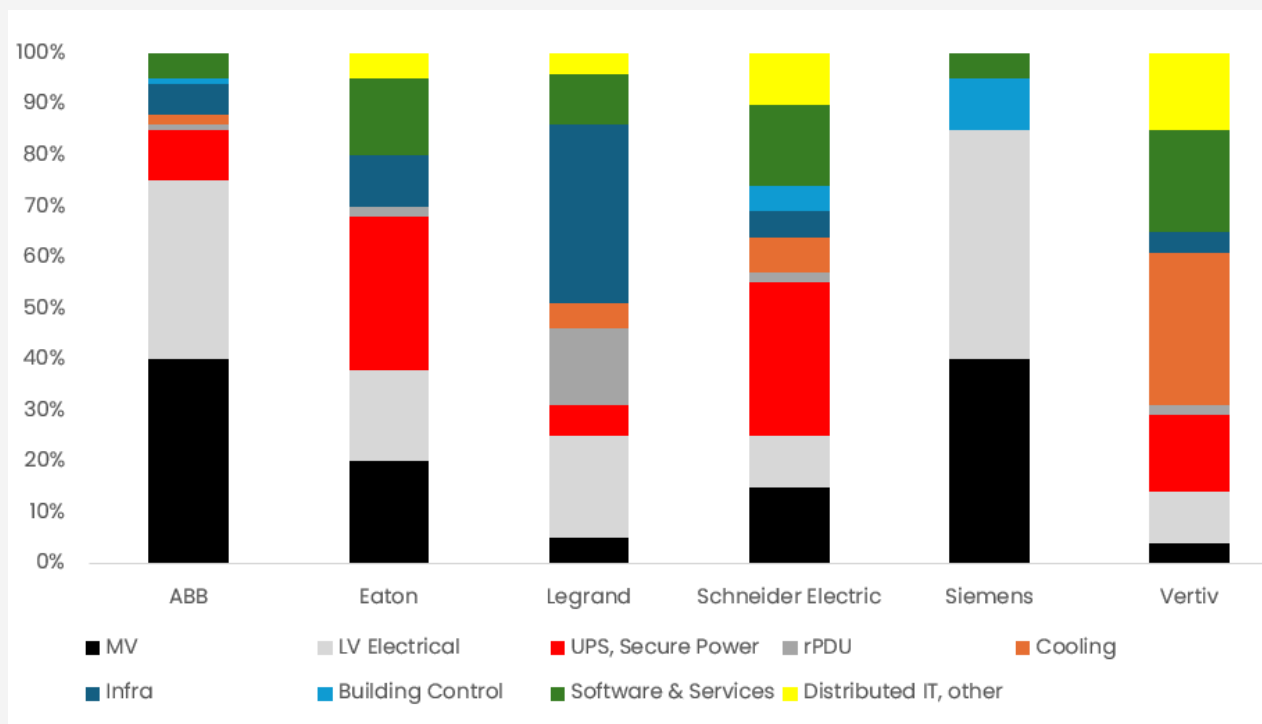
Source: Company data, Oxcap Analytics estimates. ABB excludes DC exposure in Motion (only includes Electrification). Siemens excludes DC exposure in Digital Industries (Control, Digital Twin etc). Includes distributed IT exposure. Eaton excludes Boyd Acquisition (2026 closure).

Figure 2: Share of Datacenter of 2025 group revenues



Source: Company data, Oxcap Analytics estimates. ABB excludes DC exposure in Motion (only includes Electrification). Siemens excludes DC exposure in Digital Industries (Control, Digital Twin etc). Includes distributed IT exposure.

Figure 3: Indicative estimate of DC exposure to product category (2025 revenues)



Source: Company data, Oxcap Analytics estimates based on management comments, industry reports and market share statistics.

Note: Distributed IT and other includes small UPS business. LV Electrical includes power electronics, LV transformers and busways. Schneider and Vertiv and to some extent Eaton have sizable distributed IT businesses selling secure power solutions to non-DC customers. The **Schneider** breakdown is adapted from the information provided at the Dec 2025 CMD, which is best-in-class. **Vertiv** disclosed a breakdown at its May 2026 CMD. **Siemens** is estimated based on disclosure at its Dec 2024 SI CMD which shows that DC at that time represented about 15% of each Electrification and Electrical Products revenues which we weigh by the size of the business with Electrification mostly related to MV but also some LV solutions while Electrical Products is mostly LV Solutions. **Legrand's** breakdown is estimated based on the group's communication at the Q4 2025 and Q1 2026 results calls and clarification provided by IR but as categories are bundled, our confidence in the numbers is lower. **ABB** does not provide a break-down of its datacenter exposure (despite its otherwise best-in-class disclosure approach) but commented that the biggest contributor to its data center revenues is the Smart Power division, followed by Distribution Solutions and (with distance) Smart Buildings which allows for some approximation. **Eaton** does not provide a break-down of its datacenter business and hence represents our estimate of triangulating market positions from industry research reports and company commentary. It excludes the Boyd acquisition that closed in March 2026. The Boyd acquisition adds \$1.7 bn of annualized 2026E sales, according to Eaton, with 80% of it reflecting liquid cooling. On a pro-forma basis, Cooling would then represent ~15% of 2026 datacenter sales.

800VDC Transition and impact on suppliers and market opportunity

Figure 4: Transition Risk & Opportunities – a simple summary table

Transition	Timeline	What happens?	Where is the opportunity?	Where is the risk?
Phase 1	2027+	Traditional rack PDU's phased out	HVDC sidecar ABB, Eaton, Schneider, Vertiv, Delta	Traditional Rack PDU Legrand < 20% of portfolio
Phase 2	2028+	Low voltage UPS starting to be phased out	HVDC sidecar ABB, Eaton, Schneider, Vertiv, Delta	Low Voltage UPS Eaton, Schneider, Eaton Vertiv, lesser degree ABB and Legrand
Phase 3	2028-29	HVDC sidecar replaced by battery rack Addition of AC/DC rectifier in grey space	Battery pack, AC/DC rectifiers ABB, Eaton, Schneider, Vertiv, Delta incl others	HVDC sidecar, centralized UPS ABB, Eaton, Schneider, Vertiv, Delta
Phase 4	2029+	Solid state transition, AC architecture disappears	Solid State Transformers, Power Electronics ABB, Eaton, China XD, GEV, Hitachi, Delta among others DC busways and protection ABB, Eaton, Schneider, Legrand, Vertiv and others	Traditional Low Voltage AC architecture ABB, Eaton, Schneider, Legrand, and others

Source: Oxcap Analytics, based on SemiAnalysis data

We use below the four phases as laid out by SemiAnalysis to discuss the impact. However, there is really only visibility on Phase 1, with future designs still in flux. Different customers will likely also take different paths with

customer specific designs (particularly outside the NVIDIA ecosystem) being important differentiators. Ultimately, given the road-map of the chip design in terms of power density, the ecosystem needs to move to DC with the design in Phase 4 adopting a "first principles" approach to what it should look like in theory, if we were not beholden to currently available technologies and production capacities.

- **Phase 1** (2027 onwards) of the transition is mostly additive in terms of electrical content, increasing spending per MW by 12% according to Figure 6, which is an incremental opportunity for most suppliers.
- However, **traditional Rack PDUs** are phased out. This is a small market at ~2% of electrical spend (ex generator) according to SemiAnalysis but relevant to its market leader Legrand accounting for maybe 15% of revenues (Oxcap estimate, not confirmed by the company). Legrand commented on this topic during the Q&A of the Q4 2025 results and Q1 2026 results. The company has said that the transition to 800VDC could impact about ~20% of Legrand's revenues negatively while the impact on the other ~80% of sales is neutral to positive (we believe this is in absolute terms, not relative to competitors). The ~20% "at risk" includes the company's Rack PDU business but also its UPS offering. Management sees this being offset by sizeable growth driven by an expanding power draw and size of the racks. It also mentioned the cooling opportunity for the sidecar. Legrand is introducing DC busways among other solutions targeting the 800V DC market. The company stated that it has made more than 30 acquisitions in the datacenter space and organically expanded its offering as the market evolved. It has built a datacenter business from scratch since 2017 and will continue to adapt going forward as architectures develop. Legrand will provide more information about this topic at its 2026 CMD on the 29th of September in Singapore. Schneider, Vertiv, Eaton and ABB also sell Rack PDUs but the relative share of datacenter revenues is materially less and likely more than offset by the HVDC sidecar opportunity.
- The incremental spending in Phase 1 relates to the **HVDC sidecar**, a \$0.4mn per MW opportunity (or \$11bn at the peak in 2028, according to SemiAnalysis and discussed later). We expect Schneider, Vertiv, Eaton, Delta Electronics and ABB, among others to go after this opportunity with products under development. The product consists of rectifier modules, power semiconductors, battery back-up unit and potentially supercapacitors with a mix of in-house components and sourced in components. However, this is probably only a temporary market opportunity given what happens in later design iterations.
- **In Phase 2** (2028 onwards), the rack goes 800V DC native, rather than having a power shelf, moving the conversion closer to the chip. The rest of the architecture remains in place. SemiAnalysis believes that we will also see the traditional **low voltage UPS start to lose its role** and eventually become obsolete, being replaced with batteries integrated into the DC architecture. Traditional architectures use a UPS to convert AC to DC to charge a battery and to convert the battery power back to AC to support continuous operation with steady power quality during outages. These conversion steps are no longer needed in a DC architecture. Some customers may maintain low voltage UPS, some will deploy MV UPS higher up in the power train while others will get rid of them (Meta, Google already have designs without UPS). UPS are the traditional core of the datacenter business at Schneider and Eaton but also important to Vertiv and others and represent $\frac{1}{3}$ of datacenter electrical content in current designs. Moving the UPS function into the Power Shelf/PSU or sidecar can significantly reduce spending for customers, increase efficiency and reduce the addressable market for suppliers.
- **Phase 3** (2028/2029) has a limited impact as the HVDC sidecar is replaced with a battery pack and an AC-DC rectifier sitting higher up in the stack, for a similar total addressable market and likely similar vendors.
- **Phase 4** (after 2029) represents a clean design where AC power from the grid or behind-the-meter solutions (unless it's a fuel cell) gets converted by a solid state transformer to 800V DC and fed directly to a battery pack near the rack, feeding straight through to the compute trays. The existing AC voltage infrastructure disappears. SemiAnalysis expects the cost of the solid state transformer to be substantially larger (initially at \$1.2mn per MW) than the saving on the low voltage electrical infrastructure at \$0.55mn per MW. However, the incremental opportunity may not be captured by the existing low voltage suppliers but a larger set of companies in what is an emerging field of start-ups, Asian competitors and technology developments by the existing transformer and electrical equipment industry. We could not find yet information on designs being published by Schneider, Vertiv or Legrand. Eaton has acquired a start-up while ABB has invested into DG Matrix. Transformer companies Siemens Energy, GE Vernova and Hitachi are developing solid state transformers. Solid state transformers are a completely different product to traditional transformers as they are built on electronic components and not copper and steel. Hence we could see value shift from the low voltage leaders to the diversified power companies, start-ups and suppliers making SiC MOSFETS, IGBTs and capacitors, including Infineon, Wolfspeed, STM, Mitsubishi Electric among many others. DC suppliers with a strong background in power electronics and DC technologies are relatively better positioned, including Delta Electronics and ABB.

The SemiAnalysis report table shown of \$mn per MW has an inconsistency as it shows the Power Shelf/PSU opportunity going up by 50% while in the text, the firm is also talking about a 10x increase. However, the companies we discuss have a limited exposure to this opportunity.

Company comments towards the SemiAnalysis assessment

We asked the datacenter electrical suppliers under coverage for feedback on the SemiAnalysis report.

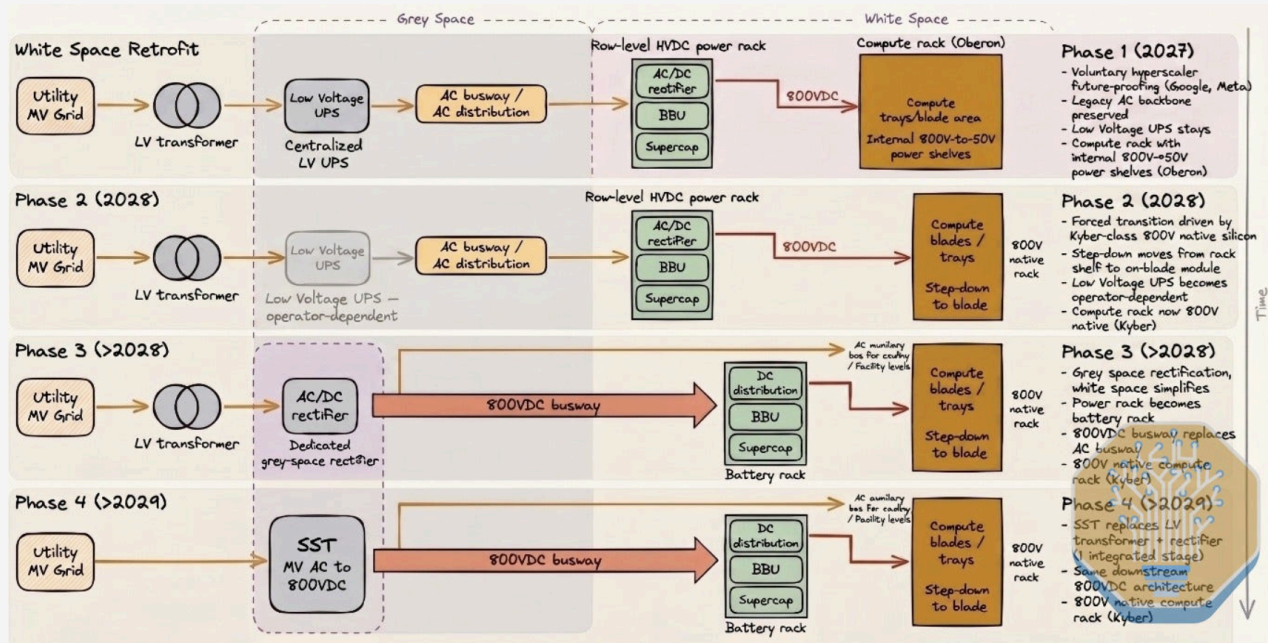
- **Legrand** commented that it will update the market with its September CMD on its strategy and positioning for 800V DC and it does not agree with the report conclusions. The company had already commented at its Q4 2025 results in February that it expects to see "tens and tens" of different architectures and that it is important to be architecture agnostic to work with all hyperscalers and co-location providers, rather than betting on a specific overall technology path.

- **Eaton** highlighted the earlier comments about an increase in its revenue opportunity per GW from \$1.2–\$1.5bn to \$1.2–2.9bn for AI datacenters, also due to the addition of battery systems for genset replacement and grid services, which could expand to \$3.4bn including the cooling opportunity post the proposed Boyd acquisition. It sees its capabilities to integrate power along the full value chain and cooling as a differentiated opportunity. It believes that acquisition of solid state transformer company Resilient Power Systems accelerates its development of DC technology by a couple of years. It said that no matter if the architectures include the sidecar, or the DC UPS / Rectifier, or the MVSST, Eaton is well positioned and will be ready to execute.
- **ABB** expects DC technology to comprise ~40–50% of capacity being installed by hyperscalers and for AI datacenters in 2030. Relative to the SemiAnalysis assessment, ABB also highlights its offering in low voltage UPS (vs the MV UPS profiled in the report). ABB also noted that in addition to the datacenter exposure in Electrification discussed, it also supplies solutions from its Motion division to datacenters, including synchronous condensers used to support transient load and drives and motors that support cooling. However it did not quantify this exposure. ABB also highlighted at its 2025 CMD its strong position in the DC technology, given its technology position and application of DC in EV-charging and Marine, among other end-markets. It introduced the first solid state circuit breaker in 2022.
- **Schneider** highlighted the announcement on May 31, 2026 by SoftBank for a major investment to accelerate AI infrastructure development in France, with a commitment to develop and operate 5 GW of AI data center capacity nationwide representing a potential investment of up to €75 billion. This includes an initial €45 billion investment to deliver around 3.1 GW. Schneider's role, as SoftBank's key industrial partner, is to operate a prefabrication facility for data center power modules to support both SoftBank's needs and broader market demand. These data center power modules will be powered by Schneider Electric low-voltage, medium-voltage, UPS, cooling and related technologies. Schneider agrees with the overall direction – 800VDC is a multi-phase transition driven by increasing rack power density and efficiency needs. In practice, adoption is likely to be more gradual and heterogeneous across operators and geographies, rather than a single step-function transition. Schneider believes that its approach of combining electrification, automation and software digitalization into a coherent full power architecture for next-generation infrastructure is an important differentiation. Developments like 800V DC or solid-state technologies are individual building blocks within a broader system transformation, where its key differentiator is the ability to design, integrate and operate a system-level solution at scale where the value is ultimately concentrated – not just individual hardware components. Schneider is fully prepared for the shift to 800V DC, which is driven by rising AI compute density rather than infrastructure preference, with racks moving beyond ~400kW. Its immediate solution is the 800V sidecar, leveraging EV standards to enable localized AC to DC conversion, delivering efficiency gains and higher power density, while positioning Schneider as an end-to-end system integrator. The transition will be gradual: sidecars from ~2027, broader distributed DC from ~2028, and more centralized architectures beyond 2029. This is an architectural alternative rather than disruption – AC remains dominant, with value reallocating toward power electronics, DC distribution and integrated solutions where Schneider is fully active.
- **Vertiv** agrees with the SemiAnalysis report in that they expect the 800V DC shift will be multiphase rather than a single switchover. On customer adoption, Vertiv expects a flexible, customer-specific path. Some will start with a sidecar, then move to dedicated DC infrastructure once silicon goes native 800V DC. Vertiv can support customers with their own offerings in MV/LV UPS across the stages, they believe that the transition could increase their TAM with some customers. The outcome of the transition is not expected to be binary – in Vertiv's experience these events are more gradual and flexible in approach than investors may be currently assuming. Their investors mainly want to understand the transition and these conversations are not new, Vertiv has been discussing it publicly for over a year. They think investor concerns are primarily rooted in seeing the outcome as binary, where Vertiv's experience suggest these things rarely are.
- **Siemens** noted that it does not see SemiAnalysis as an electrical industry research firm that they have shared information or discussed their offering with as they would with traditional industry research firms like Gartner or IDC, that have dedicated analysts that engage with companies when writing market research.

Summary of the SemiAnalysis 800V DC roadmap

- **Why switch to DC is happening:** The move to 800V DC is not simply an architectural preference; it is increasingly a physical and economic necessity. As AI racks move toward several hundred kilowatts and ultimately megawatt-class loads, the current required under traditional low-voltage architectures drives copper mass, heat losses and space requirements beyond practical limits. Because resistive losses scale with current squared, raising the distribution voltage materially reduces losses and conductor size. NVIDIA's Kyber rack architecture for Rubin Ultra is designed around this shift. SemiAnalysis states that 800V DC can transmit substantially more power through the same copper, reducing copper requirements, and improve end-to-end power efficiency by up to 5% (the very large majority of power demand relates to running the chip). At gigawatt data-center scale, that efficiency gain is economically material.
- **What does it mean for the customers:** The shift to 800V DC rewires more than the power train; it changes the strategic architecture of the data center. Higher-voltage DC affects conversion stages, protection, grounding, isolation, fault management and safety certification, while standards are still emerging. That means adoption will not follow a single path. Hyperscalers may move first in AI clusters where rack densities justify the change, while many colocation, enterprise and edge facilities are likely to retain conventional AC distribution well beyond 2030. AC will also remain relevant for much of the non-IT load, including cooling, building controls and auxiliary systems. The first phase of adoption is therefore likely to be led by the largest cloud and AI operators, with Meta and Google among the most visible early movers and Microsoft also participating, albeit potentially more selectively.

Figure 5: Four Phases of technology transition



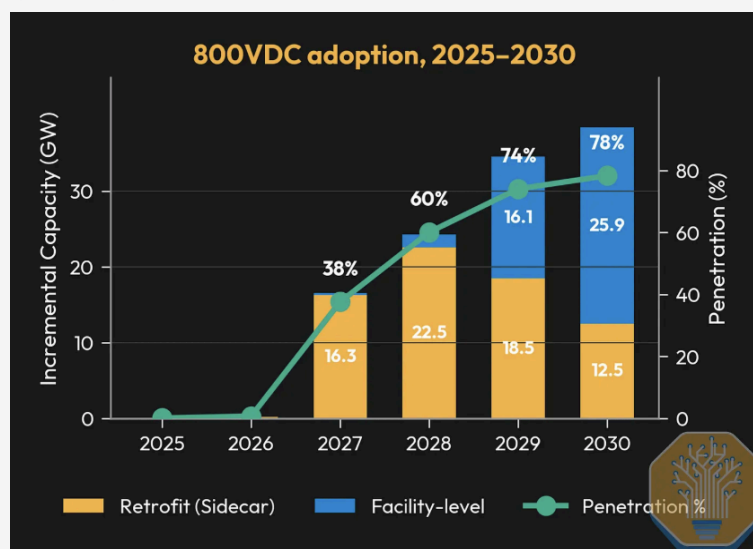
Source: SemiAnalysis, May 2026.

Figure 6: Change in market opportunity in \$ million per MW throughout the 800V DC transition

	Introduction	MV Transformers	MV Switchgear	Low voltage electrical equipment	UPS	AC-DC Rectifier	Solid State Transformer	Busway, Busbar	Rack PDU	HVDC Power Pack Sidecar	Battery Pack	Power Shelf/PSU	Total \$mn/MW	Change
Equipment spending per \$mn/MW														
Baseline \$ mn/MW	NM	0.35	0.70	0.55	1.20	0	0	0.30	0.05	0	0	0.20	3.35	
Phase 1 (The Sidecar)	2027	0.35	0.70	0.55	1.20	0	0	0.35	0	0.40	0	0.20	3.75	12%
Phase 2 (Goodbye UPS)	2028	0.35	0.70	0.55	0	0	0	0.35	0	0.40	0	0.30	2.65	-29%
Phase 3 (Centralisation)	2028/2029	0.35	0.70	0.55	0	0.20	0	0.35	0	0	0.20	0.30	2.65	0%
Phase 4 (The Desination)	>2029	0.35	0.70	0	0	0	1.25	0.25	0	0	0.20	0.30	3.05	15%

Source: SemiAnalysis, Oxcap Analytics. Excludes generator market which is included in SemiAnalysis market total as none of our companies supply them and their use is not impacted by the 800V DC transition. Generators faces challenges from trend to behind-the-meter power solutions, gas engines and battery systems which have not been reflected in the SemiAnalysis forecast of a steady market opportunity per MW for generators.

Figure 7: The real design changes start in 2029 onwards



Source: SemiAnalysis.

What Changes?

Phase 1 – Sidecar instead of Rack PDU:

Phase 1 of the 800V DC transition is generally well understood and is likely to be driven more by customer preference and future-proofing than by immediate technical necessity. The first generation of NVIDIA Vera Rubin systems should still be supportable using traditional three-phase AC power architectures, so adoption of 800V DC is likely to vary by customer, density target and deployment philosophy.

In Phase 1, most of the upstream data center electrical architecture remains familiar. Medium-voltage power is stepped down through transformers, fed into UPS systems and distributed through conventional AC infrastructure. The UPS still performs the double-conversion function: AC is converted to DC to support battery charging and energy storage, and then inverted back to AC for downstream distribution. From there, 415V AC is typically distributed via busbar to the IT rows.

The main architectural change occurs at the row or rack interface. Instead of feeding 415V AC directly into each IT rack and converting power through distributed server power supplies, the AC feed is routed into a dedicated HVDC power rack, or sidecar, located adjacent to the compute rack. This sidecar converts 415V AC to 800V DC and distributes high-voltage DC to the neighbouring IT racks.

The sidecar has three core functions: first, AC-to-DC conversion from the facility-level AC feed to 800V DC; second, integration of short-duration backup or ride-through capability through a battery backup unit; and third, transient-load management, potentially using capacitors or other energy-buffering components to handle rapid GPU load swings.

Inside the IT rack, a DC-DC power shelf then steps 800V DC down to an intermediate voltage, commonly around 50V DC, which is distributed to compute trays. Voltage regulator modules on the compute boards then perform the final conversion from the intermediate bus voltage to the sub-1V levels required by CPUs, GPUs and accelerators.

Economically, Phase 1 is an additive architecture rather than a full replacement of the existing electrical system. The traditional rack PDU may be eliminated or reduced, but the sidecar introduces new AC-DC conversion, protection, control, energy-buffering and mechanical integration content. SemiAnalysis has estimated that the sidecar / power-rack market could peak at around \$11bn in 2028, before declining as later phases move more of the DC architecture upstream.

This means Phase 1 is likely to increase, rather than reduce, the addressable market for electrical equipment. SemiAnalysis estimates the total electrical system cost could rise by around 10%, as the removal of the rack PDU is more than offset by the addition of the HVDC sidecar and associated conversion, protection and control hardware. Traditional electrical suppliers are likely to participate in this market by supplying the sidecar.

Phase 2 – Going DC native:

At this stage, the transition to 800V DC becomes less discretionary and more a function of rack density. While Phase 1 can be understood as a transitional architecture around the Oberon rack – inserting an HVDC sidecar while leaving much of the upstream AC infrastructure intact – Phase 2 is driven by the power density requirements of NVIDIA's Kyber rack design. The key architectural change is that 800V DC is distributed deeper into the compute rack, with DC-DC conversion moving closer to the compute blade. Rather than stepping 800V DC down to around 50V DC in a centralized rack-level power shelf before distribution to the trays, Phase 2 moves the step-down function closer to the load, with on-blade or near-blade power modules handling the conversion to the intermediate voltage required by the compute board.

This shift has important implications for the role of centralized low-voltage UPS systems. SemiAnalysis expects these systems to progressively lose relevance as power protection migrates closer to the rack and battery backup units assume more of the ride-through and transient-load management function. Google and Meta have already moved in this direction in parts of their infrastructure, using distributed backup architectures to improve efficiency and reduce conversion losses. However, this is unlikely to be a uniform transition. Some operators may retain centralized UPS systems for redundancy, operational familiarity or load-fluctuation management, at least through the medium term. Others may adopt medium-voltage UPS architectures at the grid or campus connection point, such as ABB's HiPerGuard.

For designs that remove the central low-voltage UPS, the electrical cost per MW can fall materially, although part of the saving is offset by higher spending on server and rack-level power electronics, including batteries. In this architecture, value shifts away from traditional UPS equipment and toward high-voltage DC distribution, rack-level energy storage, protection, controls and power supply units. The PSU and power-module opportunity is likely to accrue primarily to electronics and power-conversion suppliers such as Delta, Lite-On, Flex Power and other server power specialists, rather than only to traditional electrical equipment vendors.

Phase 3 – Centralisation:

Phase 3 is where 800V DC becomes the core electrical architecture rather than a conversion layer added near the rack. In Phases 1 and 2, AC-to-DC conversion remains close to the IT load inside an HVDC sidecar or power pack. In Phase 3, rectification moves upstream into the grey space or outdoors, creating an 800V DC backbone that feeds rows, racks and energy-buffering systems directly. Medium-voltage voltage transformers and switchgear remain essential because the utility feed and most on-site generation are still AC, although fuel cells are a partial exception because they naturally generate DC. The low-voltage AC layer, however, is materially simplified.

Much of the traditional equipment between the rectifier and the IT load is removed or reduced, including low-voltage switchgear, centralized UPS systems, floor PDUs and parts of the AC busway architecture. These functions are replaced by DC busway, battery or energy racks, DC/DC distribution units, BBU shelves and potentially supercapacitors to manage fast GPU load transients. The architecture also requires a new layer of DC protection and switching, including DC breakers, DC switchgear and solid-state circuit breakers. This creates an offsetting opportunity for suppliers: lost low-voltage AC content is partly replaced by new DC distribution, protection and control content.

Vendors are already moving in this direction. Delta has positioned a full-stack 800V DC ecosystem that includes solid-state transformers, DC busways, DC/DC power shelves and liquid cooling, while ABB is emphasizing medium-voltage protection and UPS architectures such as HiPerGuard. SemiAnalysis expects other electrical suppliers, including Legrand, to follow. The net result is not necessarily a lower electrical equipment TAM per MW, but a redistribution of value: away from the integrated HVDC power pack and conventional AC low-voltage chain, and toward separate rectifiers, battery racks, DC busway, DC protection and rack-level power electronics.

Phase 4 – Solid state transformers:

SemiAnalysis calls solid-state transformers the “holy grail” of DC power distribution because they could remove several stages of the conventional electrical chain. Instead of stepping medium voltage down through a conventional iron-and-copper transformer and then rectifying low-voltage AC into DC, an SST converts medium-voltage AC directly into low-voltage DC using high-frequency semiconductor switching, typically based on SiC MOSFETs. This can materially reduce space requirements, potentially by up to 90%, while improving efficiency, controllability and dynamic response.

The strategic attraction is that SSTs are programmable power devices rather than passive transformers. They can adjust under load, support bidirectional power flow and potentially allow datacenters to use their batteries as grid-support assets during periods of system stress. The drawback is maturity: SSTs are a new technology for mission-critical datacenter applications, with limited operating history, open questions around useful life and reliability, and additional cooling and protection requirements.

The vendor landscape includes start-ups such as DG Matrix, Amperesand, Heron Power, Hyperscale Power and Novos Power, as well as Chinese suppliers including China XD. ABB has invested in DG Matrix, while Eaton entered the market through its 2025 acquisition of Resilient Power Systems. SemiAnalysis also sees transformer incumbents such as ABB, Hitachi, Delta, Enphase and GE Vernova as potential relevant participants as the market develops (we would also add Siemens Energy).

The economics are still early to assess but potentially material. SemiAnalysis estimates that SSTs could remove roughly \$0.75m per MW of low-voltage equipment and rectifier cost, offset by an initial SST cost of \$1.0–1.5m per MW. As the technology scales, the SST cost curve should improve, making the architecture more attractive. SemiAnalysis does not expect broad adoption before 2029, but believes major hyperscalers are already running pilot projects and sees the SST market reaching around \$13bn by 2030.

Facility level impact

Even in an 800V DC architecture, data centers do not become fully DC facilities. A significant portion of the non-IT load – including chillers, pumps, fans, lighting, general HVAC and building controls – will continue to rely on AC power. CDUs are generally AC-powered today, although Delta has disclosed a DC-powered CDU and suppliers are also working on DC-based compressor technologies for chillers. The likely outcome is therefore a hybrid architecture: DC increasingly dominates the IT power train, while AC remains relevant for much of the facility infrastructure.

Backup power is also unlikely to disappear. The move to 800V DC may accelerate the use of supercapacitors, BBUs and battery energy storage for transient-load management, ride-through and short-duration backup. However, longer-duration outages remain difficult to cover economically with batteries alone, given the size and cost of the required battery overbuild relative to generator or engine based solutions. In our view, generators, fuel cells or other long-duration backup technologies will therefore remain relevant, even as short-duration resilience migrates closer to the rack.

The key investment point is that DC architectures do not eliminate the electrical equipment opportunity; they broaden it. As datacenters become a larger share of the overall electrical market, especially in the US, the incentive to develop DC-specific components increases. This should support new addressable markets in DC busway, DC protection, DC-powered cooling, rectification, BBUs, supercapacitors and rack-level power electronics.

Challenges of the transition

The NEC is the core US framework for electrical installation, but approval is ultimately down to local authorities. The 2029 NEC cycle is expected to provide a more complete treatment of 800V DC data center architectures; before then, deployments are likely to require site-specific approval and equipment-level certification. That favours hyperscalers with internal engineering resources and repeatable campus designs, while raising the adoption hurdle for smaller operators and colocation providers.

The safety challenge is that DC fault interruption is fundamentally different from AC. AC voltage crosses zero 100 or 120 times per second in 50Hz and 60Hz systems, creating natural arc-extinction points. DC has no such zero-crossing, so arcs can be more persistent and harder to interrupt. Traditional AC protection cannot simply be reused; 800V DC architectures require DC-rated protection, isolation, fault detection and potentially solid-state

breakers. This also raises the labour requirement: in a market already short of qualified electrical technicians, high-voltage DC adds another layer of training, certification and operational complexity.

What SemiAnalysis says about vendor positioning

- **ABB:** SemiAnalysis sees ABB's offering as narrower than those of Schneider Electric, Vertiv and Eaton, particularly given its limited presence in cooling. Rather than positioning itself as a one-stop-shop provider, ABB is focused on being a best-in-class supplier of discrete electrical packages, which is well-suited for hyperscalers with their own engineering teams. SemiAnalysis also highlights ABB's medium-voltage UPS offering as a potential differentiator, particularly for customers considering more centralized or upstream backup-power architectures.
- **Delta:** Delta is viewed as one of the relative winners, alongside Lite-On, given its strong position in white-space power electronics. Its offering spans power shelves, BBUs, PSUs, capacitors and related electronics, giving it exposure to the areas where value is expected to migrate under 800V DC architectures. Delta is also one of the few suppliers with a credible grid-to-chip offering that includes both power electronics and cooling, making it well positioned as the architecture shifts toward higher-density, electronics-led infrastructure.
- **Eaton:** SemiAnalysis sees Eaton as less well positioned than Vertiv and Delta in the near term, primarily because it has less exposure to white-space power electronics such as PSUs, BBUs and DC-DC converters, which are expected to become more important under 800 VDC architectures. However, SemiAnalysis does recognize Eaton's opportunity in solid-state transformers, where the acquisition of Resilient Power Systems gives the company an early position relative to much of the traditional electrical equipment peer group.
- **Legrand:** Legrand is singled out as a potential relative loser because of its high exposure to rack PDUs and row-level busway, both of which could come under pressure in Phase I of the 800V DC transition. SemiAnalysis believes that management's comments on the Q4 2025 results call understated the potential risk. While Legrand is developing DC busway solutions to adapt its portfolio, SemiAnalysis argues that its current product set offers fewer opportunities than competitors to participate in new value pools such as HVDC sidecars and advanced white-space power electronics.
- **Schneider Electric:** SemiAnalysis sees Schneider Electric as behind Delta and Vertiv in the 800V DC race, despite its strong broader datacenter franchise. It notes Schneider's leadership in medium-voltage switchgear and electrical distribution, which should remain well protected through the transition. Indeed, the move to 800V DC could make medium-voltage infrastructure more complex and therefore potentially more valuable. However, Schneider appears less advantaged than Delta or Vertiv in the white-space power electronics categories where incremental value is expected to accrue.
- **Vertiv:** SemiAnalysis highlights Vertiv's 800V DC win with Meta, alongside Delta, as evidence of the strength of its offering. It sees a near-term benefit from additional power-pack content, at least while centralized or rack-adjacent conversion remains part of the architecture. Over time, however, the potential decline in traditional UPS content could create a headwind. Like other traditional infrastructure suppliers, Vertiv is not meaningfully exposed to server-side white-space power electronics, such as PSUs and DC-DC conversion, where a growing share of incremental value may accrue.

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Alphabet Inc (GOOG.US) | Price 373.07*;
China XD Electric Co Ltd (601179.CH)*;
Delta Electronics (2308.TT) | Price TWD2,420.00*;
Eaton (ETN.US) | Price USD399.00 | Target price USD350.00 | Recommendation Rating | Equal-weight;
Enphase Energy Inc (ENPH.US) | Price USD64.00*;
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